

CSS Animation Properties

Beautiful quick-reference notes with values, meanings, and code examples

Quick idea: CSS animations are controlled by animation properties, while **@keyframes** defines what actually changes during the animation.

1. animation-name

Defines which **@keyframes** animation is applied to the element.

Value	What it does	Example
none	No animation is applied.	<code>animation-name: none;</code>
custom-name	Uses the name of a user-defined @keyframes animation.	<code>animation-name: moveBox;</code>

2. animation-duration

Sets the time required to complete one animation cycle.

Value	What it does	Example
0s	Default duration. The animation has no visible transition.	<code>animation-duration: 0s;</code>
seconds (s)	Sets duration in seconds. Larger values make the cycle slower.	<code>animation-duration: 3s;</code>
milliseconds (ms)	Sets duration in milliseconds. 1000ms equals 1 second.	<code>animation-duration: 3000ms;</code>

3. animation-delay

Sets how long the browser waits before starting the animation.

Value	What it does	Example
0s	Animation starts immediately.	<code>animation-delay: 0s;</code>
Positive time	Waits for the specified time before starting.	<code>animation-delay: 2s;</code>
Negative time	Animation starts as if it has already been running for that amount of time.	<code>animation-delay: -2s;</code>

4. animation-iteration-count

Sets how many times the animation repeats.

Value	What it does	Example
1	Runs once.	<code>animation-iteration-count: 1;</code>
2, 3, 4, ...	Runs the specified number of times.	<code>animation-iteration-count: 3;</code>

Value	What it does	Example
infinite	Repeats continuously.	<code>animation-iteration-count: infinite;</code>

5. animation-direction

Controls the direction in which each animation cycle plays.

Value	What it does	Example
normal	Plays forward from the first keyframe to the last.	<code>animation-direction: normal;</code>
reverse	Plays backward from the last keyframe to the first.	<code>animation-direction: reverse;</code>
alternate	Plays forward, then backward on the next cycle.	<code>animation-direction: alternate;</code>
alternate-reverse	Plays backward, then forward on the next cycle.	<code>animation-direction: alternate-reverse;</code>

6. animation-timing-function

Controls how the animation's speed changes during a cycle.

Value	What it does	Example
linear	Constant speed throughout.	<code>animation-timing-function: linear;</code>
ease	Slow start, faster middle, slow end. This is the default.	<code>animation-timing-function: ease;</code>
ease-in	Starts slowly and speeds up.	<code>animation-timing-function: ease-in;</code>
ease-out	Starts quickly and slows down.	<code>animation-timing-function: ease-out;</code>
ease-in-out	Starts slowly, speeds up, then slows down.	<code>animation-timing-function: ease-in-out;</code>
step-start	Jumps immediately to the next state at the start of each step.	<code>animation-timing-function: step-start;</code>
step-end	Holds the current state until the end of each step.	<code>animation-timing-function: step-end;</code>
steps(n)	Divides the animation into a fixed number of discrete steps.	<code>animation-timing-function: steps(5);</code>
cubic-bezier()	Creates a custom speed curve using four control values.	<code>animation-timing-function: cubic-bezier(0.68, -0.55, 0.27, 1.55);</code>

7. animation-fill-mode

Controls which styles are applied before the animation starts and/or after it ends.

Value	What it does	Example
none	Does not retain animation styles before or after the animation.	<code>animation-fill-mode: none;</code>
forwards	Keeps the styles from the final keyframe after the animation ends.	<code>animation-fill-mode: forwards;</code>
backwards	Applies the first keyframe during the animation delay.	<code>animation-fill-mode: backwards;</code>
both	Applies both backwards and forwards behavior.	<code>animation-fill-mode: both;</code>

8. animation-play-state

Controls whether the animation is currently playing or paused.

Value	What it does	Example
running	Animation is playing.	<code>animation-play-state: running;</code>
paused	Animation is paused at its current position.	<code>animation-play-state: paused;</code>

9. animation (Shorthand)

Combines several animation properties into one declaration.

Value	What it does	Example
Shorthand	Can specify name, duration, timing function, delay, iteration count, direction, fill mode, and play state.	<code>animation: moveBox 3s ease-in-out 2s infinite alternate;</code>

10. @keyframes

Defines the stages of an animation by specifying styles at different percentages.

Value	What it does	Example
0%	Beginning of the animation.	<code>0% { left: 0; top: 0; }</code>
25%	Quarter point of the animation.	<code>25% { left: 300px; top: 0; }</code>
50%	Halfway point.	<code>50% { left: 300px; top: 200px; }</code>
75%	Three-quarter point.	<code>75% { left: 0; top: 200px; }</code>
100%	End of the animation.	<code>100% { left: 0; top: 0; }</code>

Part A – Required Animation Example

This example combines the effects commonly required in the practical: background-color change, movement around the four sides, 2-second delay, infinite repetition, reverse direction, and a timing function.

```
animation: moveBox 8s ease-in-out 2s infinite reverse;
```

Remember

animation-name = which animation • **duration** = how long • **delay** = when it starts • **iteration-count** = how many times • **direction** = which way • **timing-function** = speed pattern • **fill-mode** = retained styles • **play-state** = running/paused